

Corona Beat - Kicking The Sedentary Habit Induced By Prolonged Social Distancing

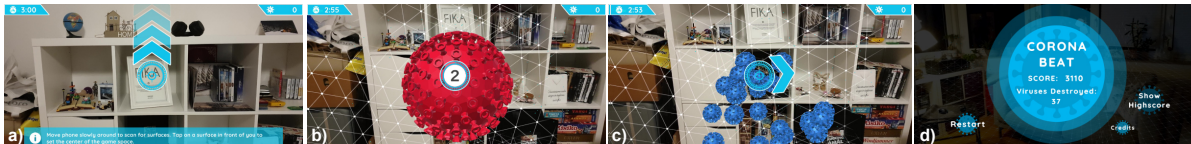
Jenny Gabel^{*†}Sukran Karaosmanoglu[‡]Celeste Mason[§]Sebastian Rings^{||}Frank Steinicke^{||}Human-Computer Interaction
Universität Hamburg

Figure 1: Illustration of the Corona Beat AR application: (a) the player starts with instructions in the AR mobile game app, (b) the player defeats a COVID-19 particle by holding the mobile phone for 5 seconds (c) the player receives feedback for a defeated COVID-19 particle, and (d) the end scene of the app shows the score and the total number of destroyed viruses.

ABSTRACT

Due to COVID-19 restrictions, most people have limited access to their typical physical activities. However, these activities play a significant role for human beings: improving cognitive, physical, and social well-being. We introduce “Corona Beat”, a proof-of-concept augmented reality customizable mobile application promoting physical activity in home environments. We designed four movement-based game scenarios that players can perform with their mobile phones: torso rotation, isometric stretching, isometric squat, and randomized rotation. We encourage these movements using COVID-19 particles, which players must defeat. In the future work, game designers/clinicians may use this work to physically train people while informing them about COVID-19.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed/augmented reality; Applied computing—Life and medical sciences—

1 INTRODUCTION

In December 2019, Corona virus disease (COVID-19) was detected, then declared a pandemic by the World Health Organization (WHO) due to its severity and rapid spread [20]. Most countries announced strict rules for daily activities such as social distancing, hygiene, and lock-down regulations to reduce the effect of this disease. Though these measures are vital, they might negatively affect physical and social activities for many people; for example, people cannot participate in gym, swim, or team sports activities [16]. Many studies reported increased sedentary behavior during COVID-19 [3, 7], which people should avoid to maintain their cognitive, physical, social well-being [5, 18]. The WHO indicates that being physically active reduces the risk of death, and improves cognitive and physical well-being compared to being physically inactive [21]. Consequently, the digital world became the center of everything: social meetings, information resources, work and education. This motivated us to

use the technology to cater to the physical activity needs of people during COVID-19.

We built the “Corona Beat” augmented reality (AR) mobile application (app) using mobile phones as the main hardware to increase accessibility and physical activity for as many people as possible. Our focus on body weight exercises maintains accessibility for a variety of people of all ages [9], especially in home environments. We incorporated a game scenario to motivate users to perform physical movements, as gamification/games can increase the enjoyment and motivation of users [2, 10], which includes physical tasks completed by defeating COVID-19 particles. This supports also the notion that we must all work together to defeat the pandemic.

2 RELATED WORK

AR mobile apps/games have become an increasingly popular entertainment and support tool for improving lives (e.g., “Pokémon Go” [13], “AR Plate Shooter” [15], and “Alien Invasion” [14]). In general, AR games’ focus on exploration and physical activity have taken the spotlight in recent years. As a result, these commercial games have also been explored in academic research. Althoff et al. [1] investigated the effect of “Pokémon Go” on physical activity using sensory wearable data; showing an increase in physical activity. Similar results also reported during COVID-19 pandemic [6], which prompted us to develop an AR app for this purpose.

Most AR fitness apps/games are meant to support healthy behaviors in a broad range of users. They often incorporate persuasive or behavior modification techniques which shows positive effects on users’ experience [4, 12]; leading us to create a customizable app. However, this requires attention to details—and creates a challenge for the development team— that might affect some user groups to differing degrees. With this in mind, we chose to focus on physical activities which can be performed by everyone, as all age groups are affected by social distancing. Hence, we aimed to create an engaging AR-based game experience for everyone who might benefit from increased motivation to maintain a more active routine.

3 CORONA BEAT: AR-BASED MOBILE APPLICATION

We present our AR-based mobile game app in three steps: technical setup, game design, and integrated physical movements.

Technical Setup The AR-based mobile app was implemented using C# in the Unity engine (v.2019.4.15f) [17] with the ARCore Development Kit [8]. We tested the app on Samsung Note 9 and One Plus 6T mobile devices with Android 10. The mobile phone served as a controller to the player’s movements in 3D space.

^{*}All authors have contributed equally to this work.

[†]e-mail: gabel@informatik.uni-hamburg.de

[‡]e-mail: karaosmanoglu@informatik.uni-hamburg.de

[§]e-mail: mason@informatik.uni-hamburg.de

^{||}e-mail: rings@informatik.uni-hamburg.de

^{||}e-mail: steinicke@informatik.uni-hamburg.de

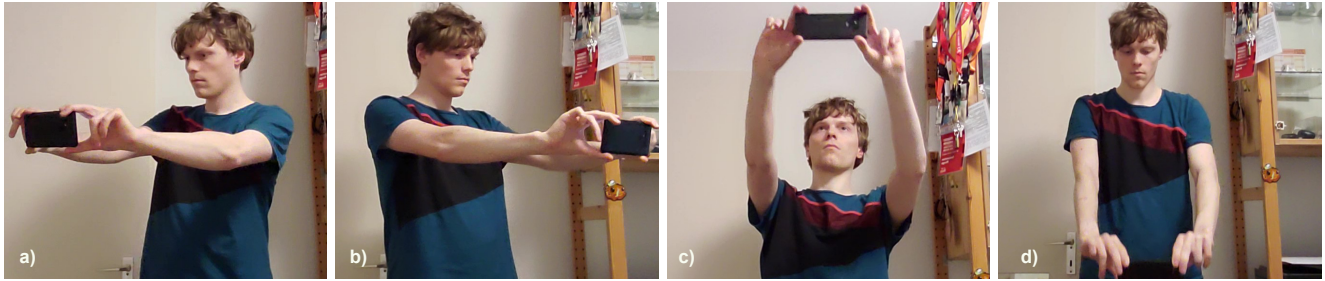


Figure 2: Different types of movements required during the games to destroy COVID-19 particles such as (a) left and (b) the right torso rotation as well as (c) up and (d) down isometric stretching movements.

Game Design The app is designed as a custom AR mobile game. The game starts with an instruction scene to describe how to perform the movements (see Fig. 1a). Following that, the player performs a series of physical movements (see Fig. 2) to beat and visually shatter 3D COVID-19 particles [11], which are spawned and anchored to the real world via ARCore anchors (see Fig. 1b,c). The spawn positions and movement patterns are partly fixed, partly randomized on the surface of a 2m spherical sector in front of the player. Also, time-based scoring was implemented to motivate players to use the app [19]. The game ends with a score board scene, which shows the total numbers of destroyed particles (see Fig. 1d).

Integrated Physical Exercise Patterns For this prototype, we focused on body weight based movements [9], which included several basic whole-body physical exercise patterns:

(i) *Torso Rotation*: To encourage this pattern, the particles were spawned at 45° angles left/right of the center viewing axis. This exercise promotes spine extension, when the targets are spawned in the vertical axis. These dynamic movements can be predicted easily and repeated quickly as a warm up (see Fig. 2a,b).

(ii) *Isometric Stretching*: For isometric patterns, the player must keep their device aimed at a COVID-19 particle for 5 seconds before targeting the next (see Fig. 2c,d). If focus is lost the exercise repeats.

(iii) *Isometric Squat*: This exercise also requires a 5 second stationary stance. The player must duck and evade particles, then stand up and target a particle at the initial location on startup.

(iv) *Randomized Rotation*: This pattern combines two movements from exercise pattern (i), but players cannot predict particle positions and must make quick motion adjustments to move between targets.

4 CONCLUSION & FUTURE WORK

We presented the “Corona Beat” AR mobile game to help reduce sedentary habits caused by COVID-19. The current version provides opportunities to customize the AR app to serve the needs of each individual. In future work, this prototype can be extended as follows: adding physical movements, personalizing the difficulty, and creating training patterns for special user groups. Further, to increase motivation to use the app, a daily reward system and multiplayer aspects (which can potentially decrease the impacts of social distancing such as depression, anxiety, and frustration [18]) could be implemented. This work can inform future game designers/developers and the healthcare community regarding possibilities for increasing physical activity using AR-based mobile game apps.

REFERENCES

- [1] T. Althoff, R. W. White, and E. Horvitz. Influence of pokémon go on physical activity: study and implications. *Journal of medical Internet research*, 18(12):e315, 2016.
- [2] A. Barnett, E. Cerin, and T. Baranowski. Active video games for youth: a systematic review. *Journal of Physical Activity and Health*, 8(5):724–737, 2011.
- [3] L. C. Bates, G. Zieff, K. Stanford, J. B. Moore, Z. Y. Kerr, E. D. Hanson, B. Barone Gibbs, C. E. Kline, and L. Stoner. Covid-19 impact on behaviors across the 24-hour day in children and adolescents: physical activity, sedentary behavior, and sleep. *Children*, 7(9):138, 2020.
- [4] D. E. Conroy, C.-H. Yang, and J. P. Maher. Behavior change techniques in top-ranked mobile apps for physical activity. *American journal of preventive medicine*, 46(6):649–652, 2014.
- [5] R. Diamond and E. Byrd. Standing up for health—improving mental wellbeing during covid-19 isolation by reducing sedentary behaviour. *Journal of affective disorders*, 277:232–234, 2020.
- [6] L. A. Ellis, M. D. Lee, K. Ijaz, J. Smith, J. Braithwaite, and K. Yin. Covid-19 as ‘game changer’ for the physical activity and mental well-being of augmented reality game players during the pandemic: Mixed methods survey study. *Journal of medical Internet research*, 22(12):e25117, 2020.
- [7] F. Gallè, E. A. Sabella, S. Ferracuti, O. De Giglio, G. Caggiano, C. Protano, F. Valeriani, E. A. Parisi, G. Valerio, G. Liguori, et al. Sedentary behaviors and physical activity of italian undergraduate students during lockdown at the time of covid-19 pandemic. *International journal of environmental research and public health*, 17(17):6171, 2020. doi: 10.3390/ijerph17176171
- [8] Google. Arcore development kit, 2018.
- [9] A. Hammami, B. Harrabi, M. Mohr, and P. Krstrup. Physical activity and coronavirus disease 2019 (covid-19): specific recommendations for home-based physical training. *Managing Sport and Leisure*, pp. 1–6, 2020. doi: 10.1080/23750472.2020.1757494
- [10] D. L. Kappen, P. Mirza-Babaei, and L. E. Nacke. Older adults’ physical activity and exergames: a systematic review. *International Journal of Human–Computer Interaction*, 35(2):140–167, 2019.
- [11] R. Kotsch. Corona virus sars-cov-2 / low mid hi poly, March 2020. License: Attribution 4.0 International (CC BY 4.0).
- [12] J. Matthews, K. T. Win, H. Oinas-Kukkonen, and M. Freeman. Persuasive technology in mobile applications promoting physical activity: a systematic review. *Journal of medical systems*, 40(3):72, 2016.
- [13] Niantic, Nintendo, and The Pokémon Company. *Pokémon Go*. Game [AR], July 2016. <https://www.pokemongo.com/en-us/>.
- [14] Param Labs. *Alien Invasion*. Game [AR], 2017. <https://apps.apple.com/us/app/alien-invasion-ar/id1263373800>.
- [15] PPCLINK Software. *AR Plate Shooter*. Game [AR]. <https://apps.apple.com/us/app/ar-plate-shooter/id1288857976>.
- [16] S. H. Shahidi, J. Stewart Williams, and F. Hassani. Physical activity during covid-19 quarantine. *Acta Paediatrica*, 109(10):2147–2148, 2020. doi: 10.1111/apa.15420
- [17] Unity Technologies. Unity real-time development platform, August 2004. San Francisco, USA. <https://unity.com/>.
- [18] A. Venkatesh and S. Edirappuli. Social distancing in covid-19: what are the mental health implications? *Bmj*, 369, 2020.
- [19] H. Wang and C.-T. Sun. Game reward systems: Gaming experiences and social meanings. In *DiGRA conference*, vol. 114. Citeseer, 2011.
- [20] W. H. O. (WHO). <https://www.who.int/news/item/27-04-2020-who-timeline-covid-19>, 2020. Accessed January 11 2020.
- [21] W. H. O. (WHO). Physical activity, 2020. Accessed January 11 2020.